

Review Diagnostic THE PRODUCT

Finding the failure nobody thought to name.

Review analytics tells you **how customers feel**. It rarely tells you **which operational failure changed**.

It discovers complaint categories from the review text itself, tests whether each one actually moved, and refuses to report anything that cannot survive that test.

■ Review Diagnostic

START

Paste a Play Store URL or package ID

com.example.grocery

Look up app

Sample Grocery App

50M+ installs

900 reviews

sample size

01 Input Paste an app, pick a sample size

■ Review Diagnostic

REPORT

Priority findings

Cold storage

PRIORITY

Freshness

WATCH

Delivery time

NO ACTION



Movement is ranked before volume.

02 Diagnostic See what moved, and by how much

■ Review Diagnostic

EVIDENCE

Cold storage

SIGNIFICANT

Confidence above threshold

Not established:

cause, failure rate or cost impact.

Next: join this lead to operational data.

03 Confidence And exactly how far to trust it

 **Discovers its own categories**
No keyword list, no taxonomy, no hypothesis to supply.

 **Ranks by movement**
The biggest category is rarely the one that changed.

 **Tests before it reports**
Corrected for testing every category at once.

 **Refuses when thin**
Says "not yet" instead of guessing.

A complaint category almost nobody was tracking rose 2.37× in a quarter, and survived every test we could put it through.

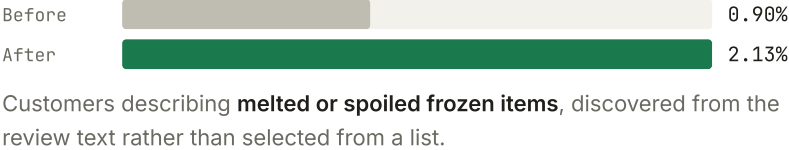
Run over **588,832 Play Store reviews** for an Indian quick-commerce app, with no hypothesis supplied. It was 20th of 25 by volume, which is why prevalence-based reporting had no reason to raise it.

What changed

SHARE OF NEGATIVE REVIEWS

2.37×

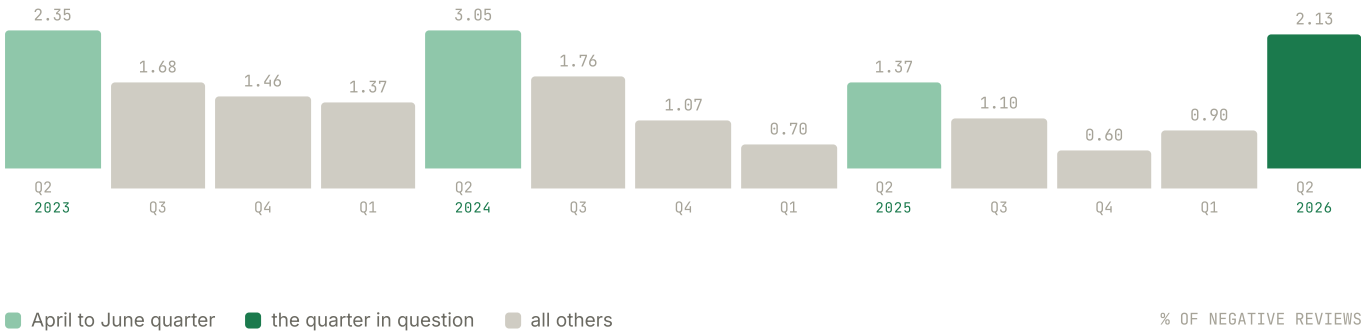
QUARTER ON QUARTER
COLD-CHAIN / MELTED ITEMS



20th
BY VOLUME
↓
1st
BY MOVEMENT

The seasonal correction

SAME DEFINITION, 13 QUARTERS BACK



The spring peak recurs in **every year on record**. Reporting 2.37× as fresh deterioration would have overstated it.

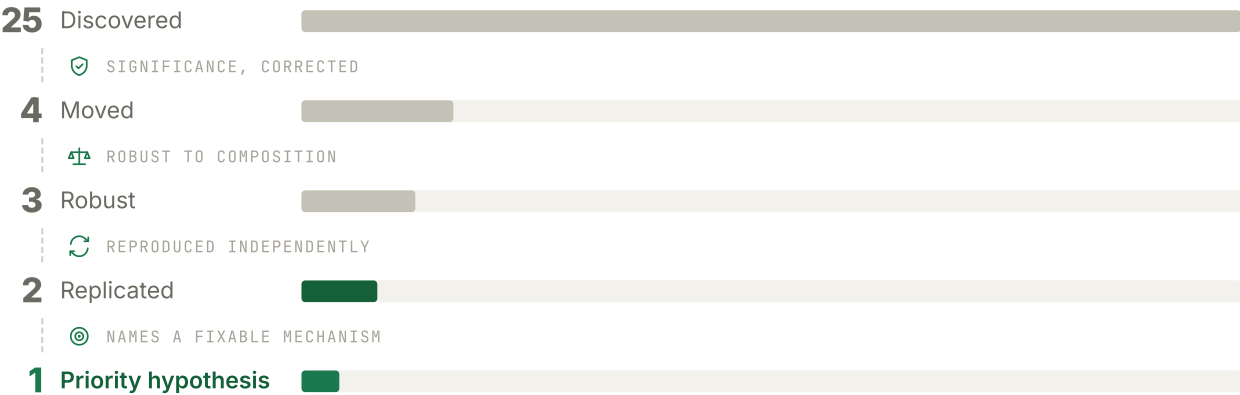
HOW MUCH WAS ACTUALLY NEW

~80% RECURRING

~20%

What survived, gate by gate

OF 25 DISCOVERED



Half the significant findings did not survive. Two of four fell at robustness and replication, which is what ranking by p-value alone would have carried into a recommendation.

What this is, and is not

READ BEFORE ACTING

OBSERVED	Complaint share rose 0.90% to 2.13% across two quarters.
VALIDATED	Significant, robust to app-version mix, reproduced independently.
HYPOTHESIS	A cold-chain failure between freezer and doorstep.
NOT ESTABLISHED	Causality, actual failure rate, or any rupee of cost.

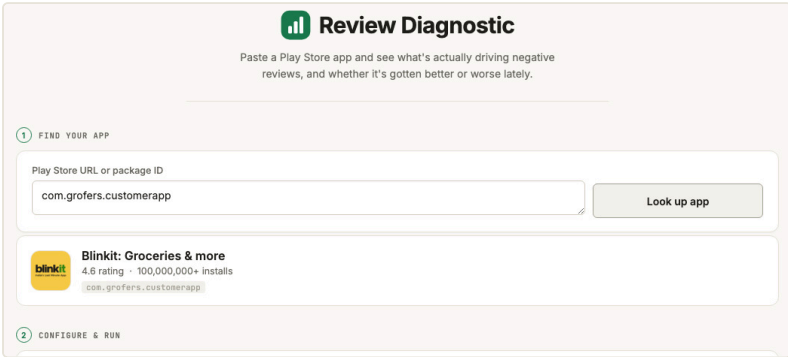
A complaint share is not an incident rate. **Direction and relative size** are informative. The underlying frequency is not recoverable.

Paste an app, get a ranked diagnostic. The hardest product decision was what the tool refuses to say.

Two inputs and one button. Everything interesting is in what happens when the data cannot support an answer, because that is where a review tool is most tempted to invent one.

One screen, two inputs

LIVE AT REVIEW-DIAGNOSTIC.ONRENDER.COM



The app is resolved and confirmed back to the user **before** any long-running work starts, so a mistyped package ID costs a second rather than a full run.

Product decisions in the interface

WHAT IS EXPOSED, AND WHAT IS NOT

No account Nothing to sign up for. The first run is the demo.

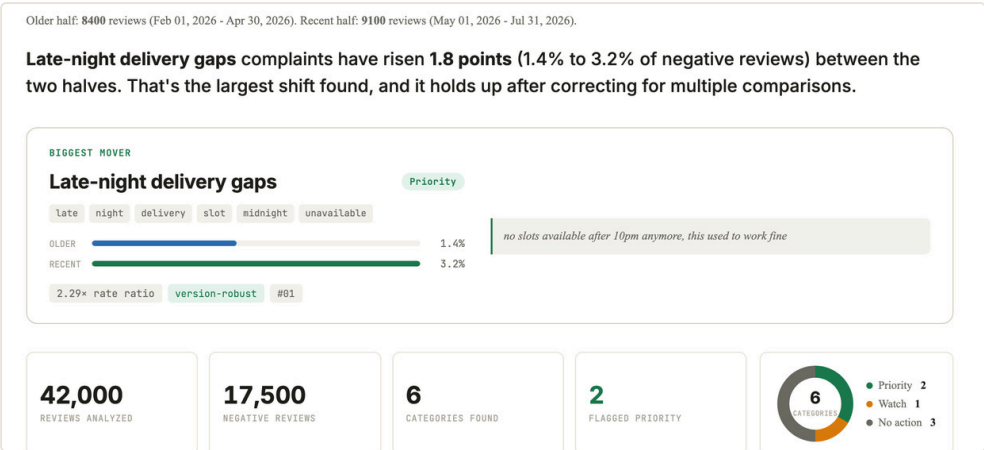
No stored data Every run is stateless, which removes the privacy question rather than answering it.

No model settings Cluster sizes and thresholds are decided by the selection rule, not exposed as knobs a user would have to guess at.

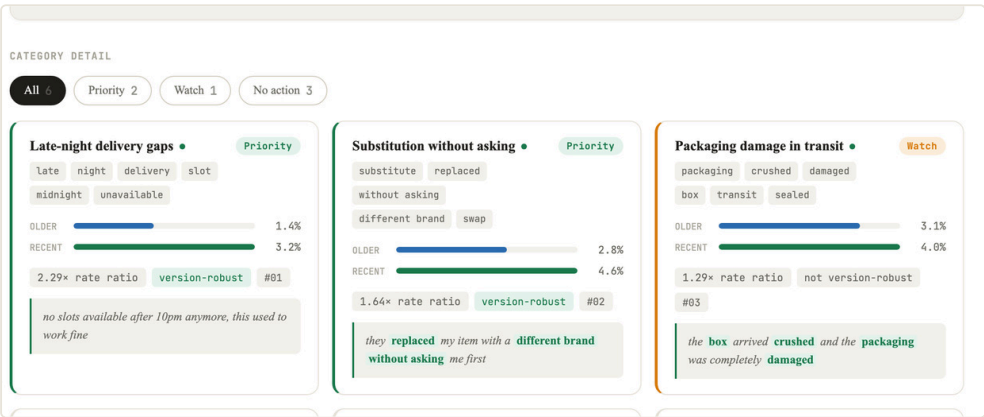
One number to choose Review count, with the instant and emailed paths labelled on the control itself.

The report

REPRESENTATIVE STATE



Headline, biggest mover, composition, then every category's shift and detail. Order mirrors the argument: **what moved, how confident, what it does not prove.**



Ranking and confidence are presented together, so weak evidence cannot masquerade as a strong finding.



Uncertainty is the interface

THE DIFFERENTIATING CHOICE

HOW CONFIDENT IS EACH FINDING?

How big each shift was, and how confident the test is that it's not just noise. The confidence bar is capped at the dashed line -- the significance threshold -- so a bar that doesn't reach it is a shift worth watching, not yet one worth trusting.

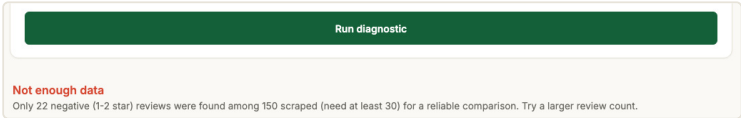
#	CATEGORY	EFFECT (K)	CONFIDENCE	ADJ. P	VERDICT
01	Late-night delivery gaps	-0.12	100%	0.000	✓ Significant
02	Substitution without asking	-0.09	100%	0.000	✓ Significant
03	Packaging damage in transit	-0.05	97%	0.027	✓ Significant
04	Refund delays	-0.03	81%	0.190	- Not yet
05	Checkout & payment errors	-0.01	27%	0.730	- Not yet

Every finding carries a confidence bar against a **dashed significance threshold**. Clear it and the row reads significant. Fall short and it reads **"not yet"**.

Most review tools present a ranked list as if every row were equally trustworthy. Here ranking and confidence are shown together, so a weak finding cannot be mistaken for a strong one.

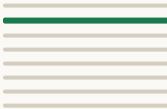
And when it cannot answer

THE REFUSAL PATH



Three separate gates can stop a run: too few reviews scraped, too few negative reviews to compare, or too few distinct patterns to form categories. Each returns a specific reason and a suggested fix.

A tool that always produces an answer is a tool that sometimes produces a fake one.

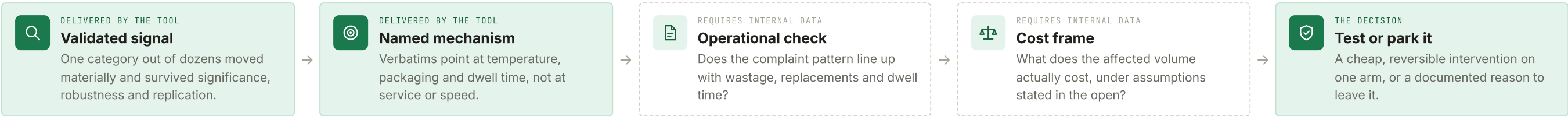


A validated signal is not yet a business case. It is a shortlist of one, and four questions worth asking.

What a finding earns is not a rollout. It earns an operations team's time, pointed at a named mechanism instead of a whole funnel.

From review signal to a decision anyone can defend

WHERE THIS TOOL'S AUTHORITY ENDS



Teams do not need to know how customers feel. They need to know which failure mode changed, and we found no lightweight way to ask that.

"Our rating is slipping" is an alarm, not a diagnosis. The operational question underneath it is narrower and harder: **of everything customers complain about, which one moved this period, and is that movement real?** We evaluated the options a small team could actually reach for. Each answers a different question well, and none of the ones we tried combined all five capabilities below.

What each option we evaluated actually answers						OUR ASSESSMENT, NOT A PRODUCT BENCHMARK
APPROACH	FINDS UNKNOWN	RANKS BY CHANGE	TESTS THE FINDING	STABLE ON RE-RUN	REFUSES WHEN THIN	BECAUSE
Star & sentiment trends rating averages, polarity						Tracks one number over time. It can tell you something broke, never which operational thing.
Fixed-taxonomy tagging keyword rules, classifiers						Bounded by its own list. A failure mode nobody wrote down lands in "other", or is averaged into a bucket that hides it.
LLM summarisation prompt over a sample						Produces a plausible story with no baseline and no test. Re-run it and the themes change.
Reading them by hand what most teams fall back on						Real texture, but unrepeatable and unscalable. Whoever reads decides what matters, and no two people rank the same list.
What we needed all five, together						No single column is novel. We found nothing lightweight that did all five together , which is the gap this was built into.
THE PATTERN						
Every option either finds things without testing them , or tests things it was told to look for . Discovery and rigour never arrive together, and a diagnosis needs both.						

Capability columns are our own assessment of these four *approaches* by construction, not a benchmark of named commercial products, several of which we did not evaluate. The claim is about what we found within reach of a small team, not about the market. The merge diagram is schematic, not measured. Section 01 shows the same effect with real figures from the validation corpus.

We tried the obvious approach first

AND IT FAILED

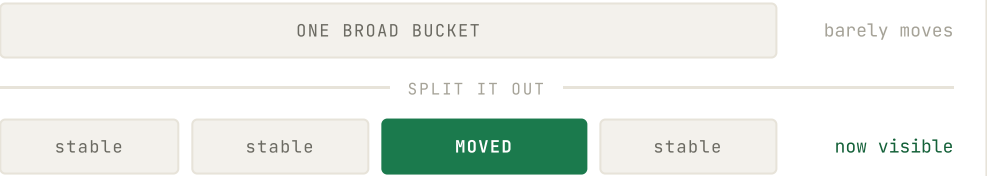
Our first build was the third row of that table: an **LDA topic model with an LLM writing up the themes**, run over reviews of a large Indian e-commerce app. It produced a readable deck. It could not survive three questions.

- "Compared to what?"**
Themes were described at a point in time, so nothing could be called better or worse.
- "Is that real or noise?"**
A theme that reads as prominent and one that genuinely rose look identical in a summary.
- "Would you get it again?"**
Re-running the topic model reshuffled the themes, and nothing flagged that it had.

The structural failure

WHY A FIXED LIST CANNOT GET THERE

A taxonomy can only separate what it was written to separate. Merge one specific failure into a broad bucket and its movement is **averaged away by everything stable around it**.



The finding only exists at the right level of detail, and **that level is not knowable before you look**. Discovery is what makes the right level available at all.

Six requirements, written before any code. Each one buys a specific kind of trust, and each one costs something real.

A capability list is easy to write and useless unless every line is paid for. These are the six the gap demanded, what each one is worth, and what we gave up to get it. Nothing here is exotic on its own. The value is that a finding has to clear all six before anyone sees it.

R1

Discover, don't assume

Categories are clustered out of the review text itself and named from their own top terms. No keyword list, no taxonomy, no hypothesis supplied.

OPEN-WORLD INPUT

BUYS

A failure mode nobody thought to look for.

COSTS

Clusters arrive unnamed, and some are linguistic artefacts.

R2

Rank by change, not size

Every category is measured as a share of that period's negative reviews in two cohorts, then ordered by the movement between them.

TWO PERIODS REQUIRED

BUYS

Separation of the chronic from the new.

COSTS

A baseline becomes mandatory, so single snapshots are useless.

R3

Test before reporting

Each shift gets a two-proportion z-test and an effect size, corrected for testing dozens of categories at once.

MULTIPLE-TEST CORRECTED

BUYS

Protection against crying wolf on noise.

COSTS

Sensitivity: real small shifts are held back as "not yet".

R4

Survive a re-run

A surviving finding is reweighted to remove composition effects, then the pipeline is executed again in a separate environment with its own clustering.

INDEPENDENT RE-RUN

BUYS

Confidence the answer is in the data, not the run.

COSTS

Two full executions, and the discipline to discard a one-run finding.

R5

Refuse when the data is thin

Sample floors at three separate stages, and a verdict that can read "**not yet**" as legibly as it reads priority.

EXPLICIT NOT YET

BUYS

The output can be trusted at face value.

COSTS

Some runs return no answer at all, which reads as failure.

R6

Zero setup to try

Paste an app, get a report. No account, no data preparation, no model settings to guess at.

NO SETTINGS TO TUNE

BUYS

The first run is the demo, with nothing to evaluate on faith.

COSTS

Hosting limits, and no way to align the comparison to your own dates.

EACH STAGE IS A FILTER, NOT A FEATURE

The trust filter

Applied in sequence, the six requirements narrow a whole corpus to the few findings worth a person's time.

Everything customers wrote

DISCOVERED, NOT ASSUMED

Categories that exist

RANKED BY CHANGE

Categories that moved

TESTED AND CORRECTED

Movements that are real

SURVIVES A RE-RUN

Findings that can be re-run

REFUSES WHEN EVIDENCE IS THIN

Priority leads worth investigating

THE TRADE, STATED PLAINLY

R1 and R2 widen what can be found. R3, R4 and R5 narrow what gets reported. A tool with only the first pair produces **confident noise**. One with only the second can only re-confirm what you already suspected.

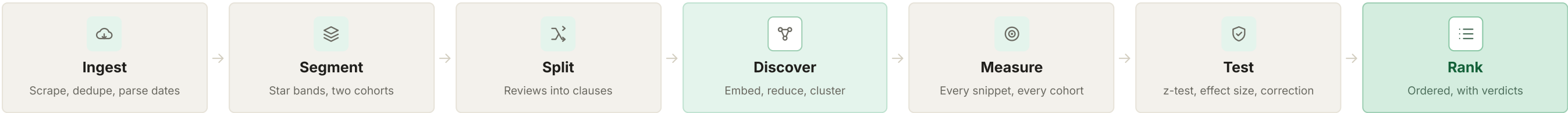
Discovery and rigour are usually sold separately. **The sequence is the product**, not any single step in it.

Seven stages, one judgement call. And that one is settled by a rule fixed before the run.

Each stage hands a stable artefact to the next. The only genuinely tunable step, how finely to cluster, is decided by a documented search rather than by looking at which setting produces the nicest answer.

The pipeline

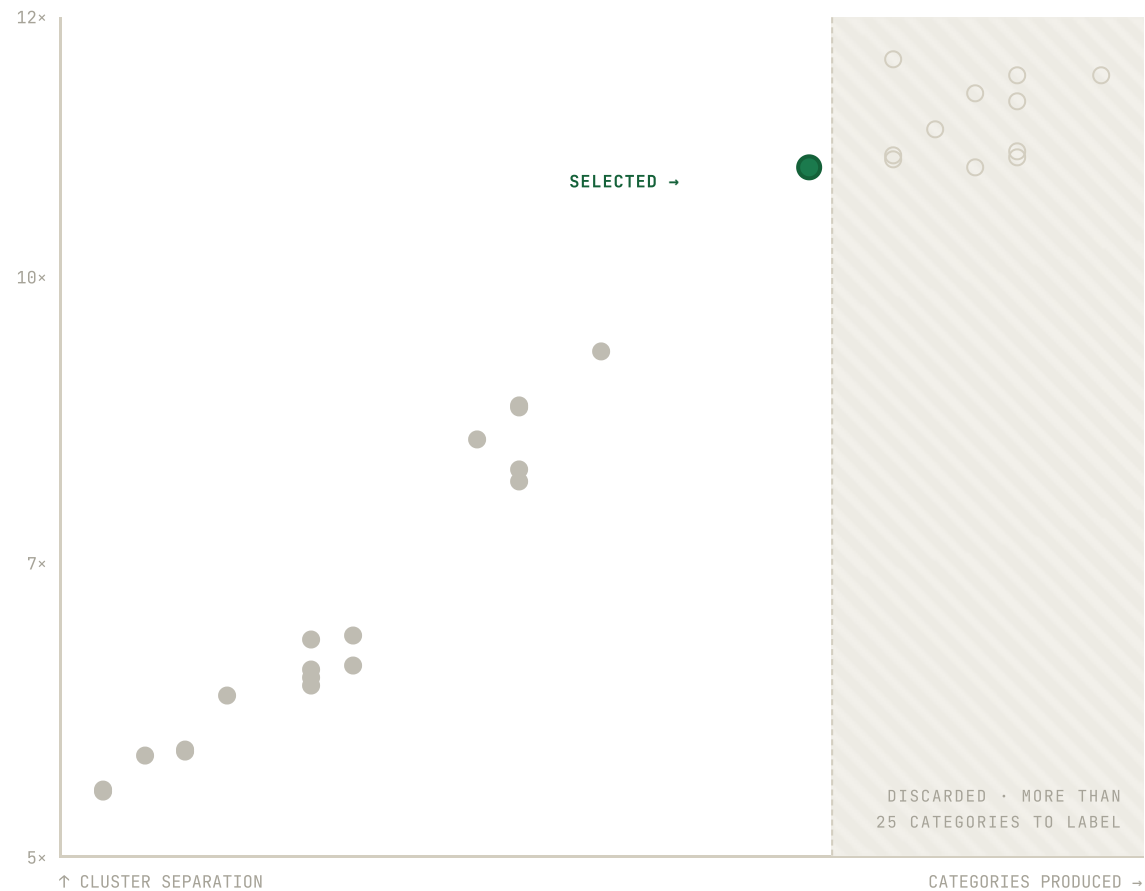
REVIEWS IN, RANKED DIAGNOSTIC OUT



The one tunable step, and how it is settled

30 CANDIDATE CONFIGURATIONS

Density clustering needs a minimum cluster size. The pipeline sweeps a grid of 30 and chooses with a rule fixed before the run, not by inspecting what each produced.



THE RULE, IN TWO STEPS

- 01 **Discard the unusable**
Anything above the category ceiling, or leaving too much text unclustered.
- 02 **Take the best of the rest**
Keep the highest separation. No inspection of which categories appeared.

Two settings separated **better**, but only by splitting the text into 27 and 29 categories. The rule preferred a labelable answer.

What the pipeline protects against

GUARDRAILS

✂

One review, two complaints
Clause splitting counts it for both instead of forcing a choice.

↗

Growth faking a trend
Shares, never counts, so more reviews cannot manufacture movement.

⚖

Findings built on a handful
A sample floor in both cohorts before any p-value is reported.

🛡

The luckiest of many draws
Correction for testing every discovered category at once.

👁

A confound hiding in the mix
Measurement never sees app version, keeping the reweighting check independent.

Chart: every configuration from the resolution search the pipeline logs on each run (negative_resolution_search.csv), here from the validation corpus in section 01. Each dot is one grid point: minimum-cluster-size as a fraction of corpus size, with minimum-samples derived per candidate. Separation ratio = mean inter-centroid distance ÷ mean intra-cluster distance. Noise never bound in this run: all 30 settings left well under the 40% ceiling, so the category range did the filtering.

Three steps to a product you could rely on, ordered by what most limits trust today.

The method is honest about its own boundary, so the product should be too. Each step below removes a specific reason a reader could discount a result.

The roadmap

ORDERED BY WHAT MOST LIMITS TRUST TODAY

1

Run it twice

Bring the replication gate into the hosted run. Report only what appears in both.

Closes the largest gap between what the method claims and what the product delivers.

2

Bring your own data

Accept uploaded tickets, NPS verbatims or reviews from any source, with user-named periods.

Removes the scraping ceiling and the store lock-in at once.

3

Watch, do not ask

Run on a schedule and alert when a category crosses the threshold.

The highest-value version is a monitor. It only works once 1 and 2 make it trustworthy.

TODAY

PRODUCT MATURITY

Deliberately not next

AND WHY

Multi-language support

Widens reach without making any single result more trustworthy.

Competitor-comparison mode

Two unreplicated runs compared is a weaker claim, not a stronger one.

An LLM narrative layer

Adds fluency over evidence, which is the failure mode this product exists to avoid.

Limits, stated plainly

NOT BURIED

LIMITATION	SEVERITY	WHAT IT DOES NOT INVALIDATE
Reviewers self-select	HIGH	Comparing two equally-biased periods mitigates it. No absolute rate should be read off the output.
Mentions are not a failure rate	HIGH	Direction and relative size are informative. The underlying frequency is not recoverable.
No cost attribution	HIGH	The tool ranks failure modes. Turning one into a figure needs internal data it has no access to.
Single hosted run	MEDIUM	A hosted finding is provisional until roadmap item 1 lands.
Small samples stay silent	MEDIUM	Correct behaviour, but it makes the free tier a demo rather than a workhorse.
One store, one language	LOW	Nothing in the method is language-bound, but nothing has been validated outside that slice.

Review Diagnostic turns a noisy review corpus into one lead worth investigating.

Not a dashboard, not a summary, and not a claim about cause. A short, tested shortlist, and an explicit account of what it would take to go further.

EXPLORE: [Live tool](#) · [Source & methodology](#)

BUILT BY

A. V. Saisantosh Patnaik

agurusantosh@gmail.com

Nithiish Makkena

nithiish.m15092004@gmail.com

 Finds what no list contains

 Proves it moved, or says it cannot

 Hands over one place to look

Independent project. The validation corpus in section 01 is also the subject of a separate 16-slide case study built from the same pipeline, which carries that finding into an economic frame under stated assumptions. The validation corpus and its figures are set out in section 01. The hosted tool scrapes live at run time and stores nothing.